

Data Warehousing and Data Mining – Complete Cheatsheet

Unit 1: Introduction to Data Warehousing

1.1 Data Lifecycle & Types

- **Data Lifecycle:** Collection → Storage → Usage → Archival → Destruction
- **Structured data:** Rows & columns, RDBMS, SQL – easiest to search
- **Semi-structured:** Has tags/markers but not rigid schema (XML, Email)
- **Unstructured:** No data model (photos, video, audio, social media)

1.2 Data Warehouse Fundamentals

- **DW:** Repository of historical data from multiple sources for decision-making
- **ETL:** Extract → Transform → Load (periodic refresh)
- **Key features (SVIN):** Subject-Oriented, Time-Variant, Integrated, Non-Volatile

Feature	Description
Subject-Oriented	Focus on subject (customer, product, sales) not operations
Time-Variant	Historical data kept; can retrieve data from months/years ago
Integrated	Heterogeneous sources combined with consistent names/types
Non-Volatile	Previous data not lost; new version inserted on update

1.3 OLTP vs OLAP

Operational Database (OLTP)	Data Warehouse (OLAP)
Current data only	Historical data
Optimized for write operations	Optimized for read/aggregation
Normalized – less redundancy	Denormalized – faster access
ER data model	Star/Snowflake model
ACID compliance	Less strict on ACID
Serves day-to-day operations	Serves decision-makers/analysts

1.4 Multidimensional Data Model

- **Data cube:** n-dimensional representation of data
- **Dimensions:** Entities to measure (e.g., time, item, location)
- **Fact table:** Contains measures (numeric facts) + foreign keys to dimension tables
- **Dimension table:** Describes dimension attributes
- **Curse of dimensionality:** Cuboid explosion; Total cuboids = $[(L_i+1)]$

1.5 Conceptual Schemas

Schema	Description	Pros	Cons
Star	Central fact table + denormalized dim tables	Simple queries, faster	Higher redundancy
Snowflake	Normalized dim tables	Saves space	More joins, slower
Fact Constellation	Multiple fact tables sharing dim tables (Galaxy)	Shared dims, flexible	Complex

1.6 OLAP Operations

Operation	Description	Example
Roll-up (aggregation)	Climbing concept hierarchy / reducing dimensions	city → country
Drill-down	Reverse of roll-up – more detail	quarter → month
Slice	Select one dimension → sub-cube	time = "Q1"
Dice	Select ≥2 dimensions → sub-cube	location="Toronto" AND time="Q1"
Pivot (rotate)	Re-orient cube view	Rotate axes

1.7 Architecture

- **Bottom tier:** DW database server + ETL tools
- **Middle tier:** OLAP server (ROLAP / MOLAP)
- **Top tier:** Front-end (query, reporting, analysis tools)
- **ROLAP:** Extended RDBMS, maps MD to relational ops
- **MOLAP:** Direct MD storage & operations (faster)

1.8 Data Marts & Components

- **Data mart:** Subset of DW for a specific dept/business line
- **Components:** Central DB, ETL tools, Metadata, Access tools
- **Metadata:** Data about data (technical + business)
- **Need for DW:** Integrate data, view summarized data, store history, help decisions, reduce analysis time

- **Trends:** Multiple data types, visualization, parallel processing, query tools, data fusion, software agents

Exam Tip: OLAP vs OLTP: OLAP = analytical (historical, denormalized, read-heavy); OLTP = operational (current, normalized, write-heavy)

Exam Tip: Star schema: denormalized dims → simpler queries, faster. Snowflake: normalized → saves space, more joins.

Unit 2: Introduction to Data Mining

2.1 KDD Process

- **KDD steps:** Cleaning → Integration → Selection → Transformation → Mining → Pattern Eval → Presentation
- **Data mining:** Essential step of KDD – extracting hidden patterns
- **Architecture:** Data Sources → DB/DW Server → Data Mining Engine → Pattern Eval → GUI. Knowledge Base guides search

2.2 Data Mining Functionalities

Functionality	Description
Concept/Class Description	Characterization (summarize target) + Discrimination (compare target vs contrasting)
Association	Find frequent itemsets (e.g., milk→bread)
Classification	Predict categorical label (supervised)
Regression	Predict continuous value (supervised)
Clustering	Group similar objects (unsupervised)
Outlier Analysis	Detect unusual data points
Evolution Analysis	Model behavior over time (time-series)

2.3 Attribute Types

Type	Properties	Examples	Operations
Nominal	Categories, no order	hair color, marital status	Mode
Ordinal	Ordered categories	size: S < M < L	Median, Mode
Interval	Numeric, equal intervals, no true zero	temp °C, calendar dates	Mean, Median, Mode
Ratio	Numeric, true zero, ratios meaningful	height, weight, Kelvin	All operations

2.4 Statistical Description

- **Central Tendency:** Mean (sensitive to outliers), Median (robust), Mode, Midrange
- **Dispersion:** Range, Quartiles (Q1, Q2/Median, Q3), IQR = Q3–Q1, Variance (σ^2), Std Dev (σ)

2.5 Issues & Applications

- **Issues:** Mining methodology, user interaction, efficiency/scalability, data diversity, security/social issues
- **Applications:** Business intelligence, market basket analysis, fraud detection, intrusion detection, customer segmentation, bioinformatics, web search, social networks

Exam Tip: Nominal: mode only. Ordinal: mode + median. Interval/Ratio: mean + median + mode.

Unit 3: Data Preprocessing

3.1 Data Quality

- **Incomplete:** Missing attribute values
- **Noisy:** Errors or outliers
- **Inconsistent:** Discrepancies in data

3.2 Data Cleaning

Handling missing values:

- Ignore tuple
- Fill manually
- Global constant ("Unknown")
- Attribute mean
- Class-mean
- Most probable value (regression/Bayesian/DT)

Smoothing noisy data:

- **Binning:** Sorted data partitioned into bins; smooth by mean/median/boundaries
- **Clustering:** Group similar points, remove outliers
- **Regression:** Linear/non-linear – fit function to data

3.3 Binning Example

```
Sorted: 4, 8, 15, 21, 21, 24, 25, 28, 34
Bin1: [4,8,15] Bin2: [21,21,24] Bin3: [25,28,34]
Smoothing:
- By mean: [9,9,9] [22,22,22] [29,29,29]
- By median: [8,8,8] [21,21,21] [28,28,28]
- By bounds: [4,4,15] [21,21,24] [25,25,34]
```

3.4 Data Integration & Transformation

- **Integration:** Combine multiple sources → coherent store. Issues: Entity identification, Redundancy detection (Chi-square/correlation)
- **Transformation:** Smoothing, Aggregation, Discretization, Normalization, Attribute Construction

3.5 Normalization

```
Min-Max:  $v' = (v - \min) / (\max - \min) \times (\text{new\_max} - \text{new\_min}) + \text{new\_min}$ 
Z-score:  $v' = (v - \mu) / \sigma$  where  $\sigma = \sqrt{(\sum (v_i - \mu)^2 / N)}$ 
```

3.6 Data Reduction

- **Dimensionality:** PCA (numeric only, finds principal components), Wavelet transforms, Attribute subset selection
- **Numerosity:** Parametric (regression, log-linear), Nonparametric (histograms, clustering, sampling)
- **Compression:** Lossless vs lossy

3.7 Discretization & Concept Hierarchy

- **Methods:** Binning, Histogram analysis, Clustering, Decision tree induction
- Replace low-level values with higher-level concepts (age: young/middle/senior)

3.8 Data Mining Primitives (DMQL)

- Task-relevant data, Kind of knowledge, Background knowledge, Interestingness measures, Visualization
- DMQL uses SQL-like syntax to specify mining tasks

Exam Tip: PCA: dimensionality reduction, numeric only, finds principal components (orthogonal vectors).

Unit 4: Data Cube Technology

4.1 Cube Materialization

- **Cuboid lattice:** Base cuboid (most detailed) $\leftrightarrow$ Apex cuboid (most generalized)
- **Full cube:** All cuboids precomputed – exponential (2^n)
- **Iceberg cube:** Only cells above min threshold
- **Closed cube:** Only closed cells (no descendant has same measure value)
- **Shell cube:** Precompute only small-dim cuboids (e.g., 3-5 dims), rest on-the-fly

Cell Type	Description
Base cell	Cell in base cuboid (all dimensions specified)
Aggregate cell	Cell from non-base cuboid (has * for aggregated dims)
Ancestor cell	i-D cell a is ancestor of j-D cell b if $i < j$ and $a_k = b_k$ wherever $a_k \neq *$
Descendant cell	Reverse of ancestor
Closed cell	No descendant with same measure value

4.2 Optimization Strategies

1. Sort, hash, group dimension attributes
2. Compute higher-level aggregates from cached lower-level results
3. Aggregate from smallest child cuboid
4. Apriori pruning for iceberg cubes

4.3 Attribute-Oriented Induction (AOI)

- **Query-oriented, on-line** generalization (vs off-line data cube)
- **Attribute removal:** Remove if large distinct values + no generalization operator
- **Attribute generalization:** Map up concept hierarchy

4.4 Class Comparison

- Compare target class vs contrasting classes
- **Steps:** Data Collection $\rightarrow$ Dimension Relevance Analysis $\rightarrow$ Synchronous Generalization $\rightarrow$ Presentation (tables, graphs, rules)

Exam Tip: For n-dim data without concept hierarchy: 2^n cuboids. With hierarchy (L_i levels each): $\prod (L_i + 1)$ cuboids

Unit 5: Mining Frequent Patterns

5.1 Basic Concepts

- **Frequent itemset:** Set of items that appear together frequently (e.g., {milk, bread})
- **Closed itemset:** No proper superset with same support
- **Support(A⇒B):** $P(A \cup B) = \text{count}(A \cup B) / |D|$
- **Confidence(A⇒B):** $P(B|A) = \text{support}(A \cup B) / \text{support}(A)$
- **Lift:** $P(A \cup B) / (P(A)P(B))$ – >1 positive corr, <1 negative corr, =1 independent

5.2 Market Basket Analysis

- Analyzes customer buying habits to find items frequently purchased together
- Helps with store layout, cross-marketing, catalog design, sales planning

5.3 Apriori Algorithm

- **Apriori property:** Any subset of a frequent itemset must be frequent
- Level-wise iterative search: $L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow \dots$ (each pass scans DB)
- **Join step:** $L_{k-1} \bowtie L_{k-1} \rightarrow C_k$ (candidate k-itemsets)
- **Prune step:** Remove candidates with infrequent subsets
- **Limitation:** Huge candidate generation, repeated DB scans

5.4 Apriori Example

```
DB with 9 transactions, min_sup=2
L1: {I1:6, I2:7, I3:6, I4:2, I5:2}
L2: {I1I2:4, I1I3:4, I1I5:2, I2I3:4, I2I4:2, I2I5:2}
L3: {I1I2I3:2, I1I2I5:2}
```

Association rules from {I1,I2,I5}:

```
I1^I2⇒I5  conf=2/4=50%  x
I1^I5⇒I2  conf=2/2=100% ✓
I2^I5⇒I1  conf=2/2=100% ✓
I1⇒I2^I5  conf=2/6=33%   x
I2⇒I1^I5  conf=2/7=29%   x
I5⇒I1^I2  conf=2/2=100% ✓
```

5.5 Improving Apriori Efficiency

- **Hash-based:** Reduce size of C_k ($k > 1$) using hash table bucketing
- **Transaction reduction:** Remove transactions without frequent k-itemsets
- **Partitioning:** Mine partitions in memory, then merge
- **Sampling:** Mine random sample (trade accuracy for speed)

5.6 FP-Growth Algorithm

- **No candidate generation** – uses FP-tree (divide & conquer)
- **Step 1:** Scan DB, find frequent 1-itemsets, sort descending
- **Step 2:** Build FP-tree: root→null, process transactions in sorted order, share common prefixes
- **Mining:** Bottom-up from leaves, construct conditional pattern base → conditional FP-tree
- Faster than Apriori for dense datasets

5.7 Types of Association Rules

Type	Description	Example
Single dimensional	Single predicate repeated	$\text{buys}(X, \text{"camera"}) \Rightarrow \text{buys}(X, \text{"printer"})$
Multidimensional	≥ 2 predicates	$\text{age}(X, \text{"20..29"}) \wedge \text{occupation}(X, \text{"student"}) \Rightarrow \text{buys}(X, \text{"laptop"})$
Multilevel	Multiple concept hierarchy levels	milk → bread (at different abstraction levels)
Quantitative	Numeric attributes	age, income

5.8 Correlation Analysis (Lift)

```
Lift(A,B) = P(A∪B) / (P(A)P(B)) = Confidence(A⇒B) / Support(B)
Lift > 1 → positive correlation
Lift < 1 → negative correlation
Lift = 1 → independent
```

Exam Tip: Apriori property: All subsets of frequent itemsets must be frequent. Used for pruning search space.

Exam Tip: FP-Growth: No candidate generation, builds compact FP-tree, divide & conquer. Faster than Apriori for dense data.

Exam Tip: Lift > 1 → positive correlation, Lift < 1 → negative correlation, Lift = 1 → independent.

Unit 6: Classification and Prediction

6.1 Fundamentals

- **Classification:** Predict categorical label (supervised)
- **Prediction/Regression:** Predict continuous value
- **Learning step:** Build model from training data
- **Testing step:** Evaluate accuracy on test data

6.2 Decision Tree (ID3)

- Top-down greedy, uses **Information Gain**
- **Entropy:** $E(D) = -\sum p_i \log_2 p_i$
- **Information Gain:** $IG(A) = E(D) - E_a(D)$
- Select attribute with highest gain as splitting criterion

Entropy = $-\sum(p_i \log_2 p_i)$
Info_Gain = Entropy(parent) - $\sum(|D_j|/|D| \times \text{Entropy}(D_j))$

Algorithm	Attribute Selection	Note
ID3	Information Gain	Favors attributes with many values
C4.5	Gain Ratio	Successor of ID3, avoids bias
CART	Gini Index	Binary splits

6.3 Naïve Bayesian Classification

- Based on **Bayes' theorem:** $P(H|X) = P(X|H)P(H) / P(X)$
- Assumes **conditional independence** of attributes
- Predict class with highest posterior: $P(C_i|X) \propto P(X|C_i)P(C_i)$

$P(C_i|X) = P(C_i) \times \prod P(x_k|C_i)$

6.4 Laplace Smoothing

- If any $P(x_k|C_i)=0$, entire product becomes 0 – add 1 to each count
- Negligible effect on large datasets, prevents zero probabilities

6.5 Backpropagation (Neural Networks)

- **Multilayer perceptron:** Input → Hidden → Output layers
- **Forward phase:** Fixed weights, propagate input to output
- **Backward phase:** Compute error, propagate back, update weights

Output: $O_j = 1 / (1 + e^{(-I_j)})$
Error: $\text{Err}_j = O_j(1-O_j)(T_j-O_j)$ [output layer]
 $\text{Err}_j = O_j(1-O_j)\sum \text{Err}_k w_{jk}$ [hidden layer]
Update: $w_{ij} += (\eta) \text{Err}_j O_i$ (η = learning rate)

6.6 Rule-Based Classification

- **IF condition THEN class** (antecedent ANDed attributes → consequent)
- **Coverage:** $n_{\text{covers}} / |D|$
- **Accuracy:** $n_{\text{correct}} / n_{\text{covers}}$
- **Conflict resolution:** Size ordering (more tests = higher priority), Rule ordering (class-based or rule-based priority list)
- **Extraction from DT:** Each root→leaf path = one rule
- **Rule simplification:** Prune conditions that don't improve accuracy

6.7 Support Vector Machine (SVM)

- Finds **maximum-margin hyperplane** (MMH) to separate classes
- **Support vectors:** Data points closest to hyperplane (determine margin)
- **Linear SVM:** For linearly separable data
- **Non-linear SVM:** Uses kernel trick (RBF, polynomial, sigmoid) to map to higher dimension
- **Advantages:** Higher speed, good with limited samples

6.8 Performance Measures

Metric	Formula
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$
Precision	$TP/(TP+FP)$
Recall	$TP/(TP+FN)$
F1-score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

- **TP:** Correct positive; **TN:** Correct negative
- **FP:** Type I error; **FN:** Type II error

6.9 Model Evaluation

- **Overfitting:** Too complex model, memorizes noise – poor generalization

- **Underfitting:** Too simple model – can't capture patterns even in training
- **k-Fold CV:** Split data into k folds; train on k-1, test on 1; repeat k times; average results
- **McNemar's test:** Chi-square on contingency table to compare 2 classifiers
- **Issues:** Data cleaning, Relevance analysis, Data normalization

Exam Tip: ID3 picks highest information gain; C4.5 uses gain ratio to avoid bias toward multi-valued attributes.

Exam Tip: Naïve Bayes assumes conditional independence – rarely true in practice but works well.

Exam Tip: Entropy = 0 when all samples same class; =1 when equally split (binary). Max uncertainty = max entropy.

Exam Tip: SVM finds max-margin hyperplane. Support vectors = closest points to decision boundary.

Exam Tip: Confusion matrix: TP/FP/TN/FN. Accuracy ≠ good metric for imbalanced data (use precision/recall/F1).

Unit 7: Cluster Analysis

7.1 Distance Measures

Euclidean: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Manhattan: $d = |x_2 - x_1| + |y_2 - y_1|$
Minkowski: $d = (|x_2 - x_1|^p + |y_2 - y_1|^p)^{1/p}$
 $p=1 \rightarrow$ Manhattan (L_1), $p=2 \rightarrow$ Euclidean (L_2)

7.2 Partitioning Methods

K-Means

- Select k random centers; assign each point to nearest; recompute centroids; repeat
- $O(n \cdot k \cdot i)$ time; sensitive to initial centers; spherical clusters only

Example: $\{(2,10), (2,5), (8,4), (5,8), (7,5), (6,4)\}$, $k=2$
Init $c_1=(2,5)$, $c_2=(6,4)$
Iter1: $C_1=\{p_1, p_2\}$ $C_2=\{p_3, p_4, p_5, p_6\}$
Iter2: $C_1=\{p_1, p_2, p_4\}$ $C_2=\{p_3, p_5, p_6\}$
Iter3: Same $\rightarrow$ Converged

K-Means++

- Smarter initialization: choose first center randomly, then pick far points with higher probability
- Rest of algorithm = standard K-means

Mini-Batch K-Means

- Uses random batches for each iteration – faster on huge datasets
- Reduces computational cost, suitable for large-scale clustering

K-Medoids (PAM)

- Uses actual data points as centers (medoids) – more robust to outliers
- Swap medoid with non-medoid if cost decreases
- Cost = $\sum |P_i - C_i|$

7.3 Hierarchical Methods

- **Agglomerative (bottom-up)**: Start with each point as cluster; merge closest pairs
- **Divisive (top-down)**: Start with one cluster; recursively split
- Distance matrix determines merging; results shown as dendrogram

7.4 Density-Based Methods

DBSCAN

- Finds arbitrary-shaped clusters; handles noise
- **Parameters**: ϵ (radius), minPts = min points for dense region
- **Core point**: $\geq$ minPts within ϵ
- **Border point**: $<$ minPts but in core's neighborhood
- **Noise**: Neither core nor border

Example: $\{(2,10), (2,5), (8,4), (5,8), (7,5), (6,4), (1,2), (4,9)\}$
 $\epsilon=2$, minPts=2
Core points: C, D, E, F, H
Noise: A, B, G
Clusters: $\{C, E, F\}$ and $\{D, H\}$

7.5 Outlier Analysis

- **Outlier**: Data point deviating from overall pattern
- Common causes: data entry errors, measurement errors, intentional, natural novelties
- Important in fraud detection, medical anomalies
- Detected via clustering (noise points), statistical tests, distance-based methods

Exam Tip: K-Means: sensitive to initial centers, works only on spherical clusters, affected by outliers. K-Medoids: more robust.

Exam Tip: DBSCAN: no need to specify k , finds arbitrary shapes, handles noise, but sensitive to ϵ and minPts.

Unit 8: Graph Mining and Social Network Analysis

8.1 Graph Mining

- Extracting frequent subgraphs/patterns from graph data
- **Applications:** chemical compounds, protein structures, social networks, web, fraud detection

8.2 Beam Search

- Heuristic search: retains only β most promising nodes at each level (beam width)
- Memory-efficient alternative to BFS/DFS/A*

```
Open = {initial state}
while Open not empty:
    n = best node from Open
    if n is goal → return path
    successors = expand(n)
    Open = Open ∪ successors
    if |Open| > β: keep best β, drop rest
```

8.3 Inductive Logic Programming (ILP)

- Intersection of ML and Logic Programming
- Derives rules (hypotheses) from positive/negative examples + background knowledge
- **Induction:** specific → general (derive rules from facts)
- **Deduction:** general → specific (derive facts from rules)

8.4 Social Network Analysis

- Study of social structures via nodes (actors) and edges (relationships)
- **Challenges:** large scale, noisy data, dynamic data
- **Link mining tasks:** Link-based classification, object type prediction, link type prediction, link existence, link cardinality, object reconciliation, group detection

8.5 Friends of Friends & Assortativity

- **FoF:** C is friend of A's friend B → indirect connection (used for friend recommendations)
- **Degree assortativity:** Tendency of nodes to connect to similar-degree nodes
- Positive coefficient = assortative (social networks), negative = disassortative
- Measured by Pearson correlation of degrees between linked nodes

8.6 Signed Networks

- Edges can be positive (friend/like) or negative (enemy/dislike)
- **Balance theory:** Balanced triples = 3 positives or 1 positive + 2 negatives. "Friend of my friend is my friend"; "Enemy of my friend is my enemy"
- **Status theory:** Positive link → higher status; Negative link → lower status
- **Conflict:** Balance and status can give opposite predictions (e.g., $A \rightarrow B \rightarrow C \rightarrow A$ triangle)

8.7 Trust Propagation

- **Atomic propagation:** Direct ($i \rightarrow j \rightarrow k \Rightarrow i$ trusts k), Co-citation, Transpose trust, Trust coupling
- **Propagation of distrust:** Trust-only, One-step distrust, Propagated distrust ($B = T - D$)
- **Iterative propagation:** Eigenvalue propagation, Weighted linear combination

Exam Tip: Balance theory: "Friend of my friend = friend; Enemy of my friend = enemy". Status theory: positive link = higher status.

Unit 9: Mining Spatial, Multimedia, Text, and Web Data

9.1 Spatial Data Mining

- Extract knowledge from spatial databases (maps, remote sensing)
- **Spatial data cube:** Nonspatial dim, Spatial-to-nonspatial dim, Spatial-to-spatial dim
- **Measures:** Numerical (revenue) or Spatial (pointers to spatial objects)
- **Application:** GIS, geo-marketing, remote sensing

9.2 Multimedia Data Mining

- Integrates image processing, computer vision, data mining
- **Description-based retrieval:** Keywords, captions, metadata
- **Content-based retrieval:** Color histogram, texture, shape, layout
- **Association types:** Image↔non-image, image↔image (non-spatial), image↔image (spatial relationships)

9.3 Text Mining & NLP

- Derive high-quality info from text documents
- **Tasks:** text classification, clustering, sentiment analysis, summarization, information extraction (IE)
- **NLP:** Understand human language – Natural Language Understanding (NLU) + Natural Language Generation (NLG)
- IE extracts structured data (entities, relationships) from unstructured text

9.4 Web Mining

Category	Description
Web Content Mining	Extracts info from web pages – text, multimedia data
Web Structure Mining	Analyzes hyperlink structure between pages
Web Usage Mining	Mines server logs for user browsing patterns

Exam Tip: Spatial data cube dimensions: nonspatial, spatial-to-nonspatial, spatial-to-spatial.

Quick Tips for Exam

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